FREE MULTIPLICATIVE CONVOLUTION AND LOG-UNIMODALITY: THREE QUESTIONS OF HASEBE, UEDA AND WANG

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Abstract. Hasebe, Ueda and Wang proved that free multiplicative convolution with the free positive multiplicative Brownian motion σ_t preserves log-unimodality of multiplicatively symmetric laws, and asked three questions about how far this extends. We answer all three. The free multiplicative convolution of two multiplicatively symmetric log-unimodal laws need not be log-unimodal: for $L \geq 6$, the logarithm of the free multiplicative square of the log-uniform law on $[e^{-L}, e^{L}]$ has a density with a strict local minimum at 0. Compact support of ν does not make $\sigma_t \boxtimes \nu$ log-unimodal for large t, since for $\nu = (\delta_{1/10} + \delta_{10})/2$ it fails for every $t > 0$; when ν is in addition log-unimodal and supported in $[\alpha, \beta]$, the convolution is log-unimodal for all $t \geq (\pi/2)(\beta/\alpha + \alpha/\beta)$. Finally, σ_t is not $\boxtimes$-strongly log-unimodal for any $t > 0$, which extends the small-time result of Hasebe, Ueda and Wang to all times. The numerical inequalities behind the first and third answers and the root count behind the second are verified in Lean 4 with Mathlib.

1. Introduction

A Borel measure μ on $\mathbb{R}$ is *unimodal* with mode a if $\mu = c\,\delta_a + f(x)\,dx$ with $c \geq 0$ and $f \geq 0$ nondecreasing on $(-\infty, a)$ and nonincreasing on (a, ∞). A probability measure μ on $\mathbb{R}_+ = (0, \infty)$ is *log-unimodal* if its image $\log_* \mu$ under $x \mapsto \log x$ is unimodal, and *multiplicatively symmetric* if it is invariant under $x \mapsto x^{-1}$. For probability measures μ, ν on $\mathbb{R}_+$ the free multiplicative convolution $\mu \boxtimes \nu$ is the law of $X^{1/2} Y X^{1/2}$ for free positive operators X, Y with laws μ, ν [12, 3]. The free positive multiplicative Brownian motion is the family $(\sigma_t)_{t>0}$ of $\boxtimes$-infinitely divisible laws with Σ-transform

$$\Sigma_{\sigma_t}(z) = \exp\Big(\frac{t}{2}\,\frac{z+1}{z-1}\Big),$$

introduced by Biane [4]. Zhong [14] described $\sigma_t \boxtimes \nu$ in detail; in particular it has a continuous density.

Hasebe, Ueda and Wang [8] studied when $\sigma_t \boxtimes \nu$ is log-unimodal. They proved that it is whenever ν is multiplicatively symmetric and log-unimodal [8, Theorem 4.2], and that it is for all $t \geq D$ when ν is supported in $[\alpha, \beta]$, $0 < \alpha < \beta$, with $\beta^4 - 3\alpha^4 < 2\alpha^3\beta$ and D an explicit constant [8, Theorem 4.4]. A probability measure μ on $\mathbb{R}_+$ is $\boxtimes$-*strongly log-unimodal* if $\mu \boxtimes \nu$ is log-unimodal for every log-unimodal ν; they proved that σ_t is not $\boxtimes$-strongly log-unimodal for every t in an interval $(0, t_0]$ [8, Theorem 4.9]. They asked the following three questions [8, Problems 4.3, 4.7 and 4.11]; numbering of [8] follows its arXiv version.

(1) "If μ and ν are multiplicatively symmetric and log-unimodal on $\mathbb{R}_+$, is the free convolution $\mu \boxtimes \nu$ log-unimodal?"

(2) "Can we delete the assumption $\beta^4 - 3\alpha^4 < 2\alpha^3\beta$ in Theorem 4.4? More precisely, if ν is a probability measure supported on $[\alpha, \beta]$ such that $0 < \alpha < \beta$, does there exist a number $D \geq 0$ (depending on ν) such that $\sigma_t \boxtimes \nu$ is log-unimodal for all $t \geq D$?"

(3) "Is it true that σ_t is not $\boxtimes$-strongly log-unimodal for any $t > 0$?"

Date: 11 October 2026.

2020 *Mathematics Subject Classification.* 46L54, 60E05.

Key words and phrases. Free multiplicative convolution, log-unimodality, free multiplicative Brownian motion, strong unimodality.

The answers are no, no, and yes. For $L > 0$ let μ_L be the law of e^U with U uniform on $[-L, L]$,

$$\mu_L(dx) = \frac{1}{2Lx}\,\mathbf{1}_{[e^{-L},e^{L}]}(x)\,dx.$$

It is multiplicatively symmetric and log-unimodal.

Theorem A. *For every $L \geq 6$ the measure $\mu_L \boxtimes \mu_L$ is not log-unimodal. More precisely, $\log_*(\mu_L \boxtimes \mu_L)$ has a real-analytic density on a neighbourhood of 0, and this density has a strict local minimum at 0.*

Theorem B. *Let $\nu = \frac{1}{2}(\delta_{1/10} + \delta_{10})$. Then $\sigma_t \boxtimes \nu$ is not log-unimodal for any $t > 0$.*

Theorem C. *For every $t > 0$ there is a log-unimodal probability measure ν_t on $\mathbb{R}_+$, given explicitly in (12), such that $\sigma_t \boxtimes \nu_t$ is not log-unimodal. Hence σ_t is not $\boxtimes$-strongly log-unimodal for any $t > 0$.*

The measure ν of Theorem B does not satisfy the hypothesis of [8, Theorem 4.4], since $10^4 - 3 \cdot 10^{-4} > 2 \cdot 10^{-3} \cdot 10$, and it is not log-unimodal. It is the failure of log-unimodality that matters: for log-unimodal ν the answer to the second question becomes positive, with a constant that depends only on the ratio β/α.

Proposition D. *Let ν be a log-unimodal probability measure supported in $[\alpha, \beta]$, $0 < \alpha < \beta$. Then $\sigma_t \boxtimes \nu$ is log-unimodal for every*

$$t \;\geq\; \frac{\pi}{2}\Big(\frac{\beta}{\alpha} + \frac{\alpha}{\beta}\Big).$$

Proposition D and [8, Theorem 4.4] do not contain each other: the former needs log-unimodality and no condition on β/α, the latter a ratio β/α close to 1 and no unimodality. Theorem B also complements [8, Theorem 4.5], where log-unimodality fails for all t because atoms of ν escape to 0 or ∞; here the support of ν is a fixed compact set.

The analogous large-time questions for the classical and free additive Brownian motions were settled by Hasebe and Ueda [6]: started from a compactly supported law μ, the free Brownian motion becomes unimodal after a time of the order of the square of the diameter of the support, while some initial laws are never unimodal; Hasebe and Sakuma [5] proved large-time unimodality of free Lévy processes whose Lévy measure has bounded support and gave an unbounded counterexample. Theorem B shows that the multiplicative analogue of the compactly supported statement fails without a unimodality assumption, and Proposition D supplies the version with one. On the unit circle, Hasebe and Ueda [7] proved that free multiplicative convolution with the free normal distributions preserves symmetric unimodality, that a symmetric law supported on an arc $\{e^{i\theta} : |\theta| \leq \varphi\}$ with $\varphi < \pi/2$ becomes unimodal after a long time, and that the convolution of the symmetric Bernoulli law on $\{1, -1\}$ with a free normal distribution is unimodal at no time; this last example is the circle counterpart of Theorem B. For recent work on the law σ_t itself see [1].

The proof of Theorem A is complex-analytic. The η-transform of $\mu_L \boxtimes \mu_L$ has an explicit inverse on $(-\infty, 0)$, and we show that this inverse inverts it on all of $\mathbb{C} \setminus [0, \infty)$. By multiplicative symmetry the η-transform maps the unit circle into itself, and its boundary value at the point 1 can then be read off from the explicit inverse. Near 1 the density of the logarithm is the imaginary part of an explicit analytic function, and the sign of its second derivative at 0 reduces to the elementary inequality (9). Theorems B and C and Proposition D use Hasebe, Ueda and Wang's criterion [8, Lemma 4.1], recalled in Lemma 2.2: log-unimodality of $\sigma_t \boxtimes \nu$ is equivalent to a bound of two on the number of roots of a family of equations. For Theorem B we exhibit four roots by the intermediate value theorem at every t. For Theorem C the witness is an atom at 1 together with a log-uniform law on a long interval $[e^{-\ell b}, 1]$, scaled with t so that a single normalized equation has four roots for all t at once.

The inequalities (9), the root count of Theorem B given the criterion, and the numerical inequalities and the choice of parameters in the proof of Theorem C are verified in Lean 4 [10] with Mathlib [11]; Section 6 lists what is and is not formalized. Numerical computations, which are not used in any proof, show the shape of the densities involved (Figures 1 and 2).

2. Preliminaries

For a probability measure λ on $\mathbb{R}_+$ let

$$\psi_\lambda(z) = \int_{\mathbb{R}_+} \frac{xz}{1-xz}\,\lambda(dx), \qquad \eta_\lambda(z) = \frac{\psi_\lambda(z)}{1+\psi_\lambda(z)}, \qquad z \in \mathbb{C}\setminus[0,\infty).$$

Both are analytic on $\mathbb{C}\setminus[0,\infty)$. For z in the upper half-plane $\mathbb{C}^+$ and $x>0$ we have $\operatorname{Im}\frac{xz}{1-xz} = \frac{x\operatorname{Im} z}{|1-xz|^2} > 0$, so $\operatorname{Im}\psi_\lambda > 0$ and $1+\psi_\lambda \neq 0$ on $\mathbb{C}^+$; since $\eta_\lambda = 1-(1+\psi_\lambda)^{-1}$,

$$\operatorname{Im}\eta_\lambda(z) = \frac{\operatorname{Im}\psi_\lambda(z)}{|1+\psi_\lambda(z)|^2} > 0 \quad (z\in\mathbb{C}^+), \qquad \eta_\lambda(z)\neq 1 \quad (z\in\mathbb{C}\setminus[0,\infty)). \tag{1}$$

On $(-\infty,0)$ the function ψ_λ takes values in $(-1,0)$, and η_λ is an increasing bijection of $(-\infty,0)$ onto itself. Its inverse defines the Σ-transform $\Sigma_\lambda(z) = \eta_\lambda^{-1}(z)/z$, and $\Sigma_{\mu\boxtimes\nu} = \Sigma_\mu\Sigma_\nu$ on $(-\infty,0)$ [12, 3]. The Cauchy transform $G_\lambda(\zeta) = \int(\zeta-x)^{-1}\lambda(dx)$ satisfies $1+\psi_\lambda(z) = z^{-1}G_\lambda(z^{-1})$, hence

$$G_\lambda(\zeta) = \frac{\zeta^{-1}}{1-\eta_\lambda(\zeta^{-1})}. \tag{2}$$

Writing λ^{-1} for the image of λ under $x\mapsto x^{-1}$, a direct computation gives $\psi_{\lambda^{-1}}(z) = -1-\psi_\lambda(1/z)$, and therefore

$$\eta_{\lambda^{-1}}(z)\,\eta_\lambda(1/z) = 1. \tag{3}$$

Finally $(\mu\boxtimes\nu)^{-1} = \mu^{-1}\boxtimes\nu^{-1}$, because $(X^{1/2}YX^{1/2})^{-1} = X^{-1/2}Y^{-1}X^{-1/2}$ [8, Section 4].

The next lemma is the form in which non-unimodality will be detected.

Lemma 2.1. *Let λ be a unimodal measure on $\mathbb{R}$ and I an open interval on which λ has a continuous density g. Then g has no strict local minimum in I.*

Proof. Let $\lambda = c\,\delta_m + f\,dx$ be as in the definition, so $f=g$ almost everywhere on I. Suppose $g(y) > g(y_0)$ for $0<|y-y_0|<\varepsilon$, with $(y_0-\varepsilon, y_0+\varepsilon)\subset I$. If $m\geq y_0$, then f is nondecreasing on $(y_0-\varepsilon,y_0)$, so $g(y)\leq g(y')$ for almost all $y<y'$ in that interval; letting $y'\uparrow y_0$ through such points and using continuity gives $g(y)\leq g(y_0)$ for almost every $y\in(y_0-\varepsilon,y_0)$, a contradiction. If $m\leq y_0$, the same argument on $(y_0,y_0+\varepsilon)$, where f is nonincreasing, gives a contradiction. □

For $R\in(0,\pi)$ put

$$\kappa_R(x) = \frac{1}{2(\cosh x - \cos R)}, \qquad x\in\mathbb{R}.$$

Since $\frac{r\xi}{1+r^2\xi^2-2r\xi\cos R} = \kappa_R(\log r+\log\xi)$, Hasebe, Ueda and Wang's criterion [8, Lemma 4.1, (1)⇔(3)] reads as follows in logarithmic coordinates. It rests on Zhong's description of $\sigma_t\boxtimes\nu$ [14].

Lemma 2.2 (Hasebe–Ueda–Wang)**.** *Let $t>0$ and let ν be a probability measure on $\mathbb{R}_+$. Then $\sigma_t\boxtimes\nu$ is log-unimodal if and only if for every $R\in(0,\pi)$ the equation*

$$\frac{\sin R}{R}\int_{\mathbb{R}_+}\kappa_R(x+\log\xi)\,\nu(d\xi) = \frac{1}{t} \tag{4}$$

has at most two solutions $x\in\mathbb{R}$.

3. Two log-uniform laws: proof of Theorem A

Fix $L>0$, put $a=e^{-L}$, $\mu=\mu_L$ and $\rho=\mu\boxtimes\mu$. The support of ρ is contained in $[a^2,a^{-2}]$, and ρ is multiplicatively symmetric because μ is.

3.1. **The inverse of η_ρ.** Integrating,

$$\psi_\mu(z) = \frac{1}{2L}\int_a^{1/a} \frac{z\,dx}{1-xz} = \frac{1}{2L}\operatorname{Log}\frac{1-az}{1-z/a},$$

where $z \mapsto (1-az)/(1-z/a)$ maps $\mathbb{C}\setminus[a,1/a]$ into $\mathbb{C}\setminus(-\infty,0]$ and Log is the principal branch. Solving $\eta_\mu(z)=w$, that is $\psi_\mu(z)=w/(1-w)$, gives $\eta_\mu^{-1}=q_L$ on $(-\infty,0)$, where

$$q_L(w) = \frac{1-E(w)}{a-E(w)/a}, \qquad E(w)=\exp\Big(\frac{2Lw}{1-w}\Big).$$

By the multiplicativity of Σ, the inverse of η_ρ on $(-\infty,0)$ is

$$Q_L(w) = \frac{q_L(w)^2}{w}. \tag{5}$$

The function q_L is meromorphic on $\mathbb{C}\setminus\{1\}$, with simple zeros where $E(w)=1$ (at 0 and at the points $i\pi k/(L+i\pi k)$, $k\in\mathbb{Z}\setminus\{0\}$, all in the open unit disc) and simple poles where $E(w)=a^2$, that is at the points $w_k = 1+iL/(\pi k)$, $k\in\mathbb{Z}\setminus\{0\}$; the point 1 is an essential singularity. Hence Q_L is meromorphic on $\mathbb{C}\setminus\{1\}$, analytic at 0 with $Q_L(0)=0$, and its poles are the w_k. Let $P=\{1\}\cup\{w_k : k\neq 0\}$, a closed set.

Lemma 3.1. *$Q_L(\eta_\rho(z)) = z$ for every $z\in\mathbb{C}\setminus[0,\infty)$.*

Proof. Let $\Omega_0=\mathbb{C}\setminus[0,\infty)$. By (1), η_ρ never takes the value 1 on Ω_0. We claim that $Z=\eta_\rho^{-1}(P)$ is a discrete closed subset of Ω_0. If $z_n\in Z$ converge to $z^*\in\Omega_0$, then $\eta_\rho(z^*)\in P$ since P is closed, and $\eta_\rho(z^*)\neq 1$, so $\eta_\rho(z^*)=w_k$ for some k. The point w_k has a neighbourhood containing no other point of P, so for large n we get $\eta_\rho(z_n)=w_k$, and as η_ρ is not constant the z_n are eventually equal to z^*. Thus $\Omega=\Omega_0\setminus Z$ is open and connected, and $Q_L\circ\eta_\rho$ is analytic on Ω. It equals the identity on $(-\infty,0)\subset\Omega$ (the points of $(-\infty,0)$ are mapped into $(-\infty,0)$, which contains no pole), hence on Ω. If Z contained a point z^*, then $\eta_\rho(z^*)=w_k$ and $|Q_L(\eta_\rho(z))|\to\infty$ as $z\to z^*$, while $Q_L(\eta_\rho(z))=z$ stays bounded. So $Z=\emptyset$ and the identity holds on Ω_0. □

3.2. **The unit circle.** By (3) with $\lambda=\rho=\rho^{-1}$, and since $\eta_\rho(\bar z)=\overline{\eta_\rho(z)}$, every z on the unit circle other than 1 satisfies $|\eta_\rho(z)|^2=\eta_\rho(z)\eta_\rho(1/z)=1$. As $\eta_\rho(-1)<0$, this gives $\eta_\rho(-1)=-1$.

On the circle the inverse Q_L is explicit. For $w=e^{i\theta}$, $0<\theta<2\pi$, write $\theta=2s$. Then $\frac{w}{1-w}=-\frac12+\frac{i}{2}\cot s$, so $E(w)=a\,e^{i\varphi}$ with $\varphi=L\cot s$, and

$$q_L(w) = -\frac{1}{B_a(e^{i\varphi})}, \qquad B_a(v)=\frac{v-a}{1-av}.$$

The Blaschke factor B_a maps the unit circle onto itself; write $B_a(e^{i\varphi})=e^{i\beta_a(\varphi)}$ with β_a continuous and $\beta_a(0)=0$. Then $\beta_a'(\varphi)=P_a(\varphi)$, where

$$P_a(\varphi)=\frac{1-a^2}{D(\varphi)}, \qquad D(\varphi)=1-2a\cos\varphi+a^2,$$

and β_a is odd, increasing, with $\beta_a(\pi)=\pi$. Consequently $Q_L(e^{i\theta})=e^{i\phi_L(\theta)}$ for $0<\theta<2\pi$, where

$$\phi_L(\theta)=2\pi-\theta-2\beta_a\big(L\cot(\theta/2)\big), \qquad \phi_L'(\theta)=L\csc^2(\theta/2)\,P_a\big(L\cot(\theta/2)\big)-1.$$

Lemma 3.2. *If $L(1-a)/(1+a)>1$, then ϕ_L is an increasing bijection of $(0,2\pi)$ onto $\mathbb{R}$, and $\phi_L(\pi)=\pi$. Its zero $\theta_*=2s_*$ satisfies $s_*\in(s_0,\pi/2)$, where $s_0=\arctan(L/\pi)$, and $\varphi_*=L\cot s_*\in(0,\pi)$.*

Proof. Since $P_a\ge(1-a)/(1+a)$ and $\csc^2\ge 1$, we get $\phi_L'\ge L(1-a)/(1+a)-1>0$. As $\theta\downarrow 0$, $L\cot(\theta/2)\to+\infty$ and $\beta_a\to+\infty$ (it increases by 2π over each period), so $\phi_L\to-\infty$; similarly $\phi_L\to+\infty$ as $\theta\uparrow 2\pi$. Clearly $\phi_L(\pi)=\pi-2\beta_a(0)=\pi$. At $\theta=2s_0$ we have $L\cot s_0=\pi$ and $\phi_L(2s_0)=2\pi-2s_0-2\pi<0$, while $\phi_L(\pi)>0$; hence $s_*\in(s_0,\pi/2)$ and $\varphi_*\in(0,L\cot s_0)=(0,\pi)$. □

Lemma 3.3. *Assume $L(1-a)/(1+a) > 1$ and put $w_* = e^{i\theta_*}$. Then $\eta_\rho(e^{i\gamma}) = e^{i\phi_L^{-1}(\gamma)}$ for $0 < \gamma < 2\pi$; in particular $\eta_\rho(e^{i\gamma}) \to w_*$ as $\gamma \downarrow 0$. Moreover Q_L is analytic at w_* with $Q_L(w_*) = 1$ and $Q_L'(w_*) = A/w_*$, where*

$$A = \phi_L'(\theta_*) = L \csc^2 s_*\, P_a(\varphi_*) - 1 > 0.$$

Proof. Since η_ρ is continuous on the circle minus 1, has modulus one there and never equals 1, we can write $\eta_\rho(e^{i\gamma}) = e^{i\vartheta(\gamma)}$ with $\vartheta : (0, 2\pi) \to (0, 2\pi)$ continuous. The poles w_k have modulus larger than one, so Lemma 3.1 gives $e^{i\phi_L(\vartheta(\gamma))} = e^{i\gamma}$. The continuous function $\phi_L(\vartheta(\gamma)) - \gamma$ takes values in $2\pi\mathbb{Z}$ and vanishes at $\gamma = \pi$, because $\vartheta(\pi) = \pi$; so $\vartheta = \phi_L^{-1}$ on $(0, 2\pi)$, and $\vartheta(\gamma) \to \phi_L^{-1}(0) = \theta_*$ as $\gamma \downarrow 0$. The point $w_* \neq 1$ lies on the circle, away from the poles, and differentiating $Q_L(e^{i\theta}) = e^{i\phi_L(\theta)}$ at θ_* gives $Q_L'(w_*) = \phi_L'(\theta_*)/w_*$. □

3.3. **The density near 1.** Assume $L(1-a)/(1+a) > 1$. By Lemma 3.3 and the inverse function theorem there are a neighbourhood U of w_* and a disc $\Delta = \{|z-1| < \delta\}$ such that Q_L maps U biholomorphically onto Δ; let $H : \Delta \to U$ be the inverse, $H(1) = w_*$. For small $\gamma > 0$ both $\eta_\rho(e^{i\gamma})$ and $H(e^{i\gamma})$ lie in U and are mapped by Q_L to $e^{i\gamma}$, so they coincide. The arc $\{e^{i\gamma}\}$ accumulates inside the connected open set $\Delta \cap \mathbb{C}^+$, where η_ρ and H are analytic, hence

$$\eta_\rho = H \quad \text{on } \Delta \cap \mathbb{C}^+. \tag{6}$$

Thus η_ρ extends continuously from $\mathbb{C}^+$ to the segment $(1-\delta, 1+\delta)$ with values $H(x)$, and by reflection from the lower half-plane with values $\overline{H(x)}$. By (2), G_ρ extends continuously from $\mathbb{C}^+$ to the real points x with $x^{-1} \in (1-\delta, 1+\delta)$, where $G_\rho(x+i0) = x^{-1}\big(1 - \overline{H(x^{-1})}\big)^{-1}$; note $H \neq 1$ near 1 because $w_* \neq 1$. By Stieltjes inversion, ρ has there the continuous density $-\pi^{-1} \operatorname{Im} G_\rho(x+i0)$, and $\log_* \rho$ has, for $|y| < \log(1+\delta)$, the density

$$g(y) = \frac{1}{\pi} \operatorname{Im} \frac{1}{1 - H(e^{-y})}. \tag{7}$$

It is real-analytic, and even because ρ is multiplicatively symmetric.

Proposition 3.4. *In this situation,*

$$g(0) = \frac{\cot s_*}{2\pi}, \qquad g'(0) = 0, \qquad g''(0) = -\frac{\csc^2 s_*}{4\pi A^3}\,(B + A\cot s_*),$$

where $B = \phi_L''(\theta_*)$. Explicitly,

$$B + A\cot s_* = -\cot s_* + \frac{L^2 a(1-a^2)\sin\varphi_*}{D(\varphi_*)^2 \sin^4 s_*}. \tag{8}$$

Proof. Let Φ be the analytic continuation of ϕ_L to a complex neighbourhood of θ_*; then $Q_L(e^{i\tau}) = e^{i\Phi(\tau)}$ there. Put $w(y) = H(e^{-y})$ and $w(y) = e^{i\tau(y)}$ with $\tau(0) = \theta_*$. From $Q_L(w(y)) = e^{-y}$ and $\Phi(\theta_*) = 0$ we get $\Phi(\tau(y)) = iy$, so $\tau'(0) = i/A$ and $\tau''(0) = -\Phi''(\theta_*)\tau'(0)^2/A = B/A^3$. Hence

$$w'(0) = i\tau'(0)w_* = -\frac{w_*}{A}, \qquad w''(0) = w_*\big(i\tau''(0) - \tau'(0)^2\big) = \frac{w_*(A+iB)}{A^3}.$$

With $F(w) = (1-w)^{-1}$ and $1 - w_* = -2i \sin s_*\, e^{is_*}$,

$$F(w_*) = \tfrac{1}{2} + \tfrac{i}{2}\cot s_*, \qquad F'(w_*) = -\frac{e^{-2is_*}}{4\sin^2 s_*}, \qquad F''(w_*) = \frac{2}{8i \sin^3 s_*\, e^{3is_*}}.$$

The first identity gives $g(0)$. Next $\pi g'(0) = \operatorname{Im} F'(w_*)w'(0) = \operatorname{Im} \frac{1}{4A\sin^2 s_*} = 0$. Finally

$$\pi g''(0) = \operatorname{Im}\big(F''(w_*)w'(0)^2 + F'(w_*)w''(0)\big) = \operatorname{Im}\Big(\frac{-ie^{is_*}}{4A^2\sin^3 s_*} - \frac{A+iB}{4A^3\sin^2 s_*}\Big) = -\frac{A\cot s_* + B}{4A^3\sin^2 s_*}.$$

For (8), differentiate $\phi_L'(\theta) = L\csc^2(\theta/2)P_a(L\cot(\theta/2)) - 1$, using $\frac{d}{d\theta}\csc^2\frac{\theta}{2} = -\csc^2\frac{\theta}{2}\cot\frac{\theta}{2}$, $\frac{d}{d\theta}L\cot\frac{\theta}{2} = -\frac{L}{2}\csc^2\frac{\theta}{2}$ and $P_a'(\varphi) = -2a(1-a^2)\sin\varphi/D(\varphi)^2$:

$$B = -L\csc^2 s_*\cot s_*\,P_a(\varphi_*) + \frac{L^2a(1-a^2)\sin\varphi_*}{D(\varphi_*)^2\sin^4 s_*},$$

and $A\cot s_* = L\csc^2 s_*\cot s_*\,P_a(\varphi_*) - \cot s_*$. □

Lemma 3.5. *For every real $L \geq 6$, with $a = e^{-L}$,*

$$a(1-a^2)(L^2+\pi^2)^2 < L(1-a)^4 \qquad \textit{and} \qquad L\,\frac{1-a}{1+a} > 1. \tag{9}$$

Proof. The second inequality is clear. For the first write $L = s+6$ with $s \geq 0$ and $T(s) = 1 + s + \frac{s^2}{2} + \frac{s^3}{6} + \frac{s^4}{24} \leq e^s$. Since $e > \frac{1359}{500}$ we have $e^6 > 400$, so $a < \frac{1}{400}$ and $(1-a)^4 \geq (\frac{399}{400})^4$; also $\pi^2 < 10$. Hence

$$a(1-a^2)(L^2+\pi^2)^2 < \frac{(L^2+10)^2}{e^6e^s} < \frac{(L^2+10)^2}{400\,T(s)},$$

and it suffices to show $(L^2+10)^2 \leq c\,L\,T(s)$ with $c = 399^4/400^3$. The difference $c(s+6)T(s) - ((s+6)^2+10)^2$ is a polynomial in s with coefficients

$$6c - 2116, \quad 7c - 1104, \quad 4c - 236, \quad \tfrac{3}{2}c - 24, \quad \tfrac{5}{12}c - 1, \quad \tfrac{1}{24}c$$

of $1, s, \ldots, s^5$, all positive because $c = 400(1 - \frac{1}{400})^4 \geq 396$ by Bernoulli's inequality. □

Proof of Theorem A. Let $L \geq 6$. By Lemma 3.5, Lemmas 3.2 and 3.3 and Proposition 3.4 apply. In (8) use $\sin\varphi_* < \varphi_* = L\cot s_*$, $D(\varphi_*) \geq (1-a)^2$, and $\sin s_* > \sin s_0 = L/\sqrt{L^2+\pi^2}$, which holds because $s_0 < s_* < \pi/2$. Since $\cot s_* > 0$,

$$B + A\cot s_* < \cot s_*(C_L - 1), \qquad C_L = \frac{a(1-a^2)(L^2+\pi^2)^2}{L(1-a)^4},$$

and $C_L < 1$ by (9). Hence $g''(0) > 0$. The density g of $\log_*\rho$ is continuous near 0, even, and has $g'(0) = 0 < g''(0)$, so it has a strict local minimum at 0. By Lemma 2.1, $\log_*\rho$ is not unimodal. □

Remark 3.6 (Numerics). For $L = 6$ one finds $s_* = 1.268996$, $w_* = -0.823297 + 0.567611\,i$, $A = 5.571866$, $B = -1.943520$, $B + A\cot s_* = -0.208942$ and $C_6 = 0.877891$, so that $g(0) = 0.049547$ and $g''(0) = 1.0544 \times 10^{-4}$. The point w_* given by Lemma 3.3 agrees to 1.6×10^{-16} with the boundary value of η_ρ at 1 obtained by solving the subordination equations of Belinschi and Bercovici [2] at $z = 1 + i\varepsilon$ and letting $\varepsilon \downarrow 0$. The whole density g is shown in Figure 1: for $L = 6$ it is supported in $[-10.307, 10.307]$, has total mass 1 to six decimals, and its maximum 0.051180 is attained at ± 6.674, so $g(0)$ is 0.968 times the maximum. A random-matrix check agrees: for $N = 1000$, the logarithms of the eigenvalues of $X^{1/2}UYU^*X^{1/2}$, with X, Y diagonal with entries e^{u_i}, u_i the midpoints of N equal cells of $[-6, 6]$, and U Haar-distributed unitary, so that X and UYU^* are asymptotically free [13], have, over 16 samples, window averages within 10^{-4} of those of g, and the same dip.

The numerics also locate the transition. The sign of $g''(0)$ changes at $L_c = 3.354645$; for $L = 2$ and $L = 3$ the computed density has a single peak at 0, while for $L = 3.5, 4, 5, 6, 8, 10, 12$ its maximum sits at $\pm 1.185, \pm 2.661, \pm 4.764, \pm 6.674, \pm 10.423, \pm 14.193, \pm 17.997$. This suggests that $\mu_L \boxtimes \mu_L$ is log-unimodal exactly when $L \leq L_c$; we prove only the case $L \geq 6$ of the failure.

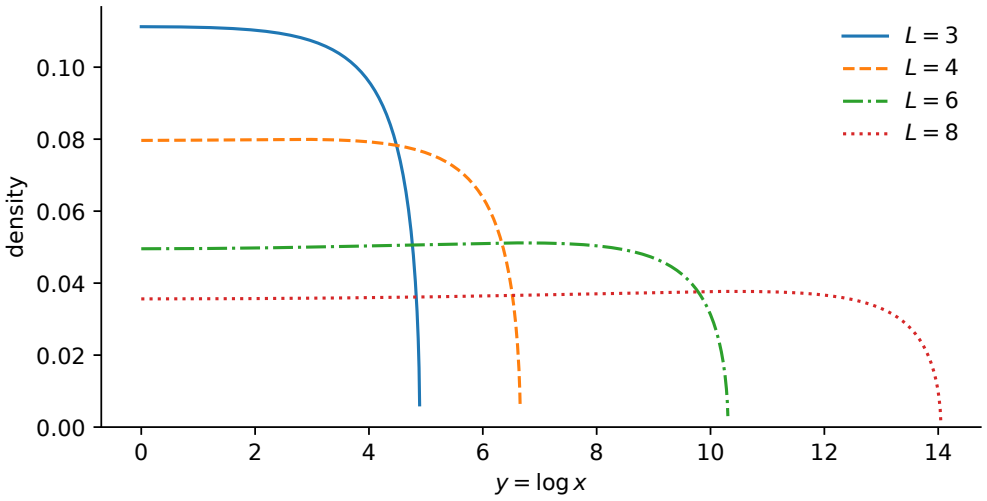


FIGURE 1. The density g of $\log_*(\mu_L \boxtimes \mu_L)$ for $L = 3, 4, 6, 8$. For $L = 3$ the density is unimodal; for $L = 4, 6, 8$ it has a local minimum at 0 (computed numerically; Theorem A proves the minimum for $L \geq 6$).

4. Two atoms: proof of Theorem B and Proposition D

4.1. **Theorem B.** Let $\nu = \frac{1}{2}(\delta_{1/10} + \delta_{10})$ and, for $R \in (0, \pi)$ and $r > 0$,

$$k_R(x) = \frac{x}{1 + x^2 - 2x\cos R}, \qquad F_R(r) = \frac{1}{2}\big(k_R(r/10) + k_R(10r)\big),$$

so that (4) with $x = \log r$ reads $\frac{\sin R}{R} F_R(r) = \frac{1}{t}$.

Lemma 4.1. *Let $R \in (0, \pi)$ and $\frac{1}{8} \leq \lambda \leq \frac{1}{4 - 4\cos R}$. Then $F_R(r) = \lambda$ has a solution in each of the intervals $(\frac{1}{1000}, \frac{1}{10})$, $(\frac{1}{10}, 1)$, $(1, 10)$ and $(10, 1000)$.*

Proof. Since $k_R(1/x) = k_R(x)$, we have $F_R(1/r) = F_R(r)$. Using $k_R(x) \leq x/(1-x)^2$ for $0 < x < 1$, $F_R(\frac{1}{1000}) = F_R(1000) \leq \frac{1}{2}(\frac{10^{-4}}{(1-10^{-4})^2} + \frac{10^{-2}}{(1-10^{-2})^2}) < \frac{1}{8}$. Next $F_R(\frac{1}{10}) = F_R(10) > \frac{1}{2}k_R(1) = \frac{1}{4 - 4\cos R}$, and $F_R(1) = k_R(10) = \frac{1}{10.1 - 2\cos R} \leq \frac{10}{81} < \frac{1}{8}$. The function F_R is continuous, so the intermediate value theorem gives the four solutions. □

Proof of Theorem B. Fix $t > 0$. If $t \leq 8$, take $R = \sqrt{t}/2 \in (0, \sqrt{2}]$. The level $\lambda = \frac{R}{t \sin R} = \frac{1}{4R \sin R}$ satisfies $\lambda \geq \frac{1}{8}$ because $R \sin R \leq \sqrt{2}$, and $\lambda \leq \frac{1}{4 - 4\cos R}$ because $1 - \cos R \leq R \sin R$, which is $\tan \frac{R}{2} \leq R$ and holds for $R \leq \sqrt{2}$. If $t > 8$, take $R \in (0, \pi)$ with $\frac{\sin R}{R} = \frac{8}{t}$, which exists because $\frac{\sin R}{R}$ decreases from 1 to 0; then $\lambda = \frac{1}{8} \leq \frac{1}{4 - 4\cos R}$. In both cases Lemma 4.1 gives four solutions of $\frac{\sin R}{R} F_R(r) = \frac{1}{t}$, so $\sigma_t \boxtimes \nu$ is not log-unimodal by Lemma 2.2. □

Remark 4.2 (The shape of the density). Zhong's description [14] shows how the failure looks. The set $V_{t,\nu} = \{r > 0 : \int \frac{r\xi}{(1-r\xi)^2} \nu(d\xi) > \frac{1}{t}\}$ is mapped by a homeomorphism onto the set of reciprocals of the points where the density of $\sigma_t \boxtimes \nu$ is positive, and the support is the closure of that set. Here the integral equals $\frac{1}{2}(\frac{1}{4}\operatorname{csch}^2 \frac{u - \log 10}{2} + \frac{1}{4}\operatorname{csch}^2 \frac{u + \log 10}{2})$ with $u = \log r$; it is convex in u on $(-\log 10, \log 10)$, increasing in r on $(0, \frac{1}{10})$, decreasing on $(10, \infty)$, and takes the value $\frac{10}{81}$ at $r = 1$. So for $t < \frac{81}{10}$ the support of $\sigma_t \boxtimes \nu$ consists of two intervals, and the logarithmic density vanishes on an interval $(-Y_t, Y_t)$; numerically $Y_t = 1.999, 1.421, 0.749, 0.333, 0.106, 0.001$ for $t = 0.05, 0.5, 2, 4, 6, 8$. For $t \geq \frac{81}{10}$ the support is an interval; at $t = \frac{81}{10}$ the density of the logarithm vanishes only at 0, and

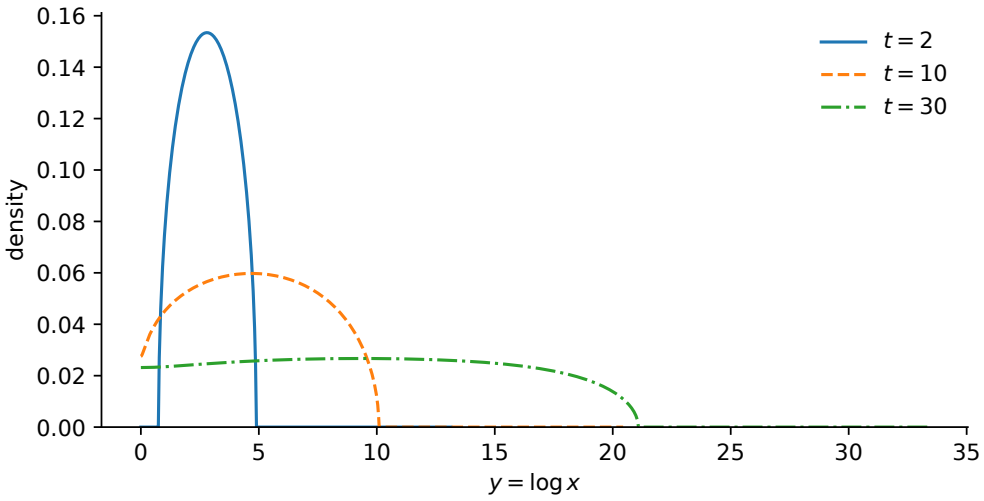


FIGURE 2. The density of $\log_*(\sigma_t \boxtimes \nu)$ for $\nu = \frac{1}{2}(\delta_{1/10} + \delta_{10})$ and $t = 2, 10, 30$, computed numerically from Zhong's formula. For $t = 2$ the support has a gap around 0; for $t = 10$ and $t = 30$ the density is positive at 0 but below its maximum.

for larger t the failure is a dip: the density of the logarithm at 0 is $0.460, 0.721, 0.868, 0.921, 0.959$ times its maximum for $t = 10, 5\pi, 30, 50, 100$ (Figure 2). For a symmetric unimodal density the value at 0 is the maximum, so these ratios show the failure directly, and they tend to 1 slowly as t grows.

4.2. **Proposition D.**

Proof of Proposition D. Let $m = \log_* \nu$, supported in $[a_1, a_2] = [\log \alpha, \log \beta]$, and $w = a_2 - a_1$, so that $\frac{\pi}{2}(\frac{\beta}{\alpha} + \frac{\alpha}{\beta}) = \pi \cosh w$. Fix $t \geq \pi \cosh w$ and $R \in (0, \pi)$, and let

$$h(x) = \int \kappa_R(x + s)\, m(ds), \qquad x \in \mathbb{R}.$$

By Lemma 2.2 we must show that $\frac{\sin R}{R}\, h(x) = \frac{1}{t}$ has at most two solutions. The function h extends analytically to a strip around $\mathbb{R}$, because m has compact support, and $h(x) \to 0$ as $|x| \to \infty$; so h is not constant on any interval.

Suppose $R \geq \pi/2$, and write $c = -\cos R \geq 0$. Then $(\log(\cosh x + c))'' = \frac{1 + c \cosh x}{(\cosh x + c)^2} > 0$, so κ_R is log-concave, and since $\int \kappa_R = \frac{\pi - R}{\sin R}$ the function $\frac{\sin R}{\pi - R}\kappa_R$ is a log-concave probability density. The image of m under $s \mapsto -s$ is unimodal, and by Ibragimov's theorem [9] its convolution with $\frac{\sin R}{\pi - R}\kappa_R$ is unimodal; this convolution is $\frac{\sin R}{\pi - R}\, h$. A unimodal continuous function that is not constant on any interval is strictly increasing to the left of its mode and strictly decreasing to the right, so each level is attained at most twice.

Suppose $R < \pi/2$. For $x \leq -a_2$ every $x + s$ with $s \in [a_1, a_2]$ is ≤ 0, where κ_R is strictly increasing; so h is strictly increasing on $(-\infty, -a_2]$, and likewise strictly decreasing on $[-a_1, \infty)$. For $x \in [-a_2, -a_1]$ we have $|x + s| \leq w$, hence $h(x) \geq \kappa_R(w)$ and

$$\frac{\sin R}{R}\, h(x) \;\geq\; \frac{\sin R}{2R(\cosh w - \cos R)} = \frac{1}{f(R)}, \qquad f(R) = \frac{2R(\cosh w - 1)}{\sin R} + 2R \tan \frac{R}{2}.$$

Both terms of f increase on $(0, \pi/2)$, so $f(R) < f(\pi/2) = \pi \cosh w \leq t$. Thus there is no solution in $[-a_2, -a_1]$ and at most one on each side. □

The case $R < \pi/2$ uses only the support of ν. For the measure of Theorem B it excludes a middle root at every $R < \pi/2$ once $t \geq \pi\cosh(\log 100) \approx 157$, and indeed the angle used in the proof of Theorem B exceeds $\pi/2$ for every $t > 4\pi$. At such angles the root count is governed by the case $R \geq \pi/2$, which is where log-unimodality of ν enters.

5. All times: proof of Theorem C

For $0 < b < \pi$ and $v \in \mathbb{R}$ let

$$K_b(v) = \frac{b\sin b}{2\pi(\cosh(bv) - \cos b)} = \frac{b\sin b}{\pi}\,\kappa_b(bv), \qquad K_0(v) = \frac{1}{\pi(1+v^2)}.$$

Lemma 5.1. *For fixed v, $b \mapsto K_b(v)$ is nonincreasing on $(0,\pi)$, and $K_b(v) \leq K_0(v)$. Moreover*

$$\int_u^v K_b = \frac{\Phi_b(v) - \Phi_b(u)}{\pi}, \qquad \Phi_b(v) = \arctan\Big(\cot\frac{b}{2}\,\tanh\frac{bv}{2}\Big). \tag{10}$$

Proof. Since $1 - \cos b = \sin b \tan\frac{b}{2}$,

$$\frac{1}{2\pi K_b(v)} = \frac{\cosh(bv) - \cos b}{b\sin b} = v^2\,\frac{\cosh(bv)-1}{(bv)^2}\cdot\frac{b}{\sin b} + \frac{\tan(b/2)}{b}$$

(the first term read as 0 when $v = 0$). Each factor is nonnegative and nondecreasing in $b \in (0,\pi)$: $(\cosh x - 1)/x^2$ is even and nondecreasing in $x \geq 0$, because $x\sinh x - 2(\cosh x - 1)$ vanishes at 0 with derivative $x\cosh x - \sinh x \geq 0$; $b/\sin b$ has derivative $(\sin b - b\cos b)/\sin^2 b > 0$; and $\tan x/x$ increases on $(0,\pi/2)$. So $K_b(v)$ is nonincreasing in b, and letting $b \downarrow 0$ gives $K_b \leq K_0$. For (10), multiply numerator and denominator of $\Phi_b'(v)$ by $\sin^2\frac{b}{2}\cosh^2\frac{bv}{2}$ and use $\sin^2\frac{b}{2} + \sinh^2\frac{bv}{2} = \frac{1}{2}(\cosh(bv) - \cos b)$; this gives $\Phi_b' = \pi K_b$. □

For $b \in (0,\pi)$, $\ell > 0$ and $p \in (0,1)$ let $\nu_{b,\ell,p}$ be the probability measure on $\mathbb{R}_+$ with

$$\log_*\nu_{b,\ell,p} = p\,\delta_0 + \frac{1-p}{\ell b}\,\mathbf{1}_{[-\ell b,0]}(s)\,ds. \tag{11}$$

It is log-unimodal with mode 1. Substituting $s = b(v-y)$, the left side of (4) at $R = b$ and $x = by$ equals $\frac{\pi}{b^2}S(y)$, where

$$S(y) = p\,K_b(y) + \frac{1-p}{\ell}\int_{y-\ell}^{y} K_b(v)\,dv.$$

Lemma 5.2. *Let $0 < b \leq \frac{1}{10}$, $\ell \geq 100$, $p = \frac{8}{5\ell}$, and $W = \ell S$. Then*

$$W(0) > \tfrac{19}{20}, \qquad W(3) < \tfrac{19}{20}, \qquad W(\ell/2) > \tfrac{19}{20}.$$

Proof. Here $W(y) = \frac{8}{5}K_b(y) + (1 - \frac{8}{5\ell})\int_{y-\ell}^{y} K_b$ and $1 - \frac{8}{5\ell} \geq \frac{123}{125}$. We use $\cot\frac{1}{20} > \frac{799}{40}$, from $\cos x > 1 - \frac{x^2}{2}$ and $\sin x < x$; $\tanh 5 > \frac{9}{10}$ and $\tanh\frac{5}{2} > \frac{29}{30}$, equivalent to $e^{10} > 19$ and $e^5 > 59$; and $\arctan x > \frac{\pi}{2} - \frac{1}{x}$ for $x > 0$. Write $X = \cot\frac{1}{20}$.

At $y = 0$, Lemma 5.1 with $K_b \geq K_{1/10}$ and $\ell \geq 100$ gives, using $K_{1/10}(0) = \frac{X}{20\pi}$ and (10),

$$W(0) \geq \frac{2X}{25\pi} + \frac{123}{125\pi}\arctan(X\tanh 5).$$

Since $X\tanh 5 > \frac{7191}{400}$, the right side exceeds $\frac{19}{20}$ as soon as $\frac{799}{500} - \frac{123}{125}\cdot\frac{400}{7191} > \frac{229}{500}\pi$, which holds because $\pi < \frac{63}{20}$.

At $y = 3$, Lemma 5.1 with $K_b \leq K_0$ gives

$$W(3) \leq \frac{8}{5}K_0(3) + \int_{-\infty}^{3} K_0 = 1 - \frac{\arctan(1/3)}{\pi} + \frac{4}{25\pi},$$

which is less than $\frac{19}{20}$ because $\arctan\frac{1}{3} > \frac{1}{3} - \frac{1}{81} = \frac{26}{81} > \frac{11}{70} + \frac{4}{25} > \frac{\pi}{20} + \frac{4}{25}$, the last step by $\pi < \frac{22}{7}$.

At $y = \ell/2$, drop the first term; since $[-50, 50] \subset [-\ell/2, \ell/2]$,

$$W(\ell/2) \geq \frac{123}{125} \int_{-50}^{50} K_{1/10} = \frac{246}{125\pi} \arctan\Big(X \tanh \frac{5}{2}\Big).$$

Since $X \tanh \frac{5}{2} > \frac{23171}{1200}$, this exceeds $\frac{19}{20}$ as soon as $\frac{17}{500}\pi > \frac{246}{125} \cdot \frac{1200}{23171}$, which holds because $\pi > 3$. □

Proof of Theorem C. Fix $t > 0$ and set

$$\ell = \max\{100,\, 95\pi t\}, \qquad b = \sqrt{\frac{19\pi t}{20\,\ell}}, \qquad p = \frac{8}{5\ell},$$

and let

$$\nu_t = \nu_{b,\ell,p}. \tag{12}$$

Since $\ell \geq 95\pi t$ we have $b^2 \leq \frac{1}{100}$, so Lemma 5.2 applies; also $p \leq \frac{2}{125}$. As $|y| \to \infty$, $W(y) \to 0$. Hence $W = \frac{19}{20}$ has a solution in each of $(-\infty, 0)$, $(0, 3)$, $(3, \ell/2)$ and $(\ell/2, \infty)$. At such a solution $S(y) = \frac{19}{20\ell} = \frac{b^2}{\pi t}$, so the left side of (4) at $R = b$ and $x = by$ equals $\frac{1}{t}$. Equation (4) thus has four solutions, and $\sigma_t \boxtimes \nu_t$ is not log-unimodal by Lemma 2.2. □

The support of ν_t is $[e^{-\ell b}, 1]$, and $\ell b = \sqrt{19\pi t\ell/20} \geq \frac{19}{2}\pi t$. So Proposition D would require $t \geq \pi \cosh(\ell b)$, far beyond t; the two results are consistent, and the witnesses must spread out on the logarithmic scale as t grows.

6. Lean verification and computations

The file `LogUnimodality.lean` in the repository listed below checks the following parts of the proofs in Lean 4 [10] with Mathlib [11] (toolchain v4.33.1). Every statement is about explicit real numbers or real functions, and the only axioms used are the standard ones (`propext`, `Classical.choice`, `Quot.sound`).

- `thmA_core`: both inequalities (9) for every real $L \geq 6$.
- `k_exp`, `four_roots_of_window`, `level_small`, `level_large` and `thmB_four_roots`: the identity $k_R(e^u) = \kappa_R(u)$, Lemma 4.1 (with the four roots ordered rather than located), the two choices of R in the proof of Theorem B, and, for every $t > 0$, the existence of $R \in (0, \pi)$ and of four positive numbers $r_1 < r_2 < r_3 < r_4$ with $\frac{\sin R}{R} F_R(r_i) = \frac{1}{t}$. Together with Lemma 2.2 this is Theorem B. `thm44_hypothesis_fails` checks that ν violates the hypothesis of [8, Theorem 4.4].
- `K_B_zero`, `arctan_third`, `crossing_zero`, `crossing_three`, `crossing_mid` and `parameters`: the value $K_{1/10}(0) = \frac{X}{20\pi}$, the bound $\arctan\frac{1}{3} > \frac{\pi}{20} + \frac{4}{25}$, the three numerical inequalities in the proof of Lemma 5.2, and the properties of ℓ, b, p used in the proof of Theorem C.

Not formalized are the free-probability input (the transforms, Lemma 2.2, the branch and boundary arguments of Section 3), Lemma 2.1, Lemma 3.2, Proposition 3.4 and the estimate of $B + A\cot s_*$ through C_L in the proof of Theorem A, Lemma 5.1 and its use in the proof of Lemma 5.2, and Proposition D. An independent check of the rational inequalities of Lemma 5.2 in exact arithmetic is included. The scripts behind Remarks 3.6 and 4.2 and the figures compute the densities from Zhong's formula and from the subordination equations; their outputs are in the repository.

Acknowledgements and funding

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel

Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25). Computational resources were provided in part by Google Cloud through its education credits program (credit 530550103).

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. The log-uniform counterexample of Theorem A was first found in September 2026 by GPT-5.6 Sol, an OpenAI model, running in scheduled ChatGPT tasks, and the witness of Theorem C in September 2026 by an OpenAI GPT model. The two-atom measure of Theorem B was found in September 2026 by an OpenAI Codex agent for $t \geq 5\pi$ and checked by a second Codex agent. In October 2026, Claude Opus 5.5 (Anthropic), working in Claude Code, rewrote and checked the proofs, extended Theorem B to all $t > 0$, proved Proposition D, carried out the numerical computations and the Lean 4 verification with Mathlib, searched the literature, and prepared the manuscript; a separate Claude Opus 5.5 instance reviewed it. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The scripts for the computations of Remarks 3.6 and 4.2 and the figures, an exact-arithmetic check of the rational inequalities, and the Lean verification are available at `https://github.com/yspennstate/free-multiplicative-log-unimodality`.

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